OpusEdge: Telemetry-Guided Dynamic Compute Allocation for Dense, Mixture-of-Experts, and Hybrid SSM–Attention Architectures

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Abstract

Deploying large language models on resource-constrained hardware remains impractical due to the quadratic cost of attention and the memory demands of key–value (KV) caches, yet existing optimization pipelines treat dense, hybrid SSM–attention, and sparse mixture-of-experts (MoE) architectures as disjoint problems requiring separate solutions. We present OpusEdge, a unified telemetry-guided framework that exploits per-token importance signals—the native SSM selectivity parameter Δ ($\mathcal{O}(1)$ cost in hybrid models), the RMS hidden-state drift Δ_{proxy} ($\mathcal{O}(L)$ cost in dense models), and the router-gated importance signal IR (MoE models)—to orchestrate ten composable inference primitives and four novel architectural stabilizers across all three architecture families without any gradient retraining. Empirically, SelKV achieves a 100.5× quality advantage over random eviction at 87.5 % KV cache compression on Falcon-H1-0.5B; DenseEvic achieves 13.7× on Qwen2.5-0.5B via Δ_{proxy}; SMSA yields 3.56–4.98× measured attention speedup at 2 048 tokens; and StateCompress delivers 37.5 % hidden-state compression with near-zero perplexity degradation, while the native Δ and Δ_{proxy} signals achieve mean Spearman ρ=0.51 and ρ=0.28 correlation with token importance across 24 diverse tasks. All experiments are reproducible on a single NVIDIA RTX 4060 (8 GB VRAM); code is available at github.com/Mr-DS-ML-85/OpusEdge.

Keywords: State-space models, KV cache compression, dense transformers, hybrid architectures, mixture of experts, efficient inference, attention routing, dynamic rank, Delta-AR, SelKV.

1 Introduction

Large language models have demonstrated strong capabilities across reasoning, code generation, and instruction following, yet deploying them on consumer or mobile hardware remains challenging. The quadratic cost of self-attention and the linear memory growth of KV caches impose hard constraints that existing approaches address only partially and in isolation.

The inference ecosystem has fragmented into three competing paradigms: pure dense transformers [10, 12], which achieve high fidelity at $\mathcal{O}(S^2)$ memory and compute; hybrid SSM–attention models [4, 9], which amortize long-range dependencies through bounded state spaces; and sparse MoE models [6, 11], which activate only a subset of parameters per token. Existing optimizations—H_2O [15], SAGE-KV [2], Ada-KV [14], vLLM [7]—treat these paradigms as disjoint and require separate pipelines for each.

We reject this fragmentation. The central observation of this work is that all three architecture families emit cheap per-token diagnostic signals that carry meaningful information about token importance. In hybrid models the SSM forward pass produces the selectivity parameter Δ at zero extra cost; in dense models a normalized hidden-state drift serves as a computationally analogous proxy; and in MoE models the router softmax logits reveal expert-selection confidence. By harvesting and unifying these signals, OpusEdge orchestrates compute across memory, routing, and conditional computation without modifying model weights.

Contributions

1. **Unified signal framework.** We formalize native Δ, Δ_{proxy}, and router-gated importance IR as a unified token importance hierarchy applicable across all three architecture families, and empirically validate their correlation with attention importance (ρ=0.51 for native Δ, ρ=0.28 for Δ_{proxy} across 24 diverse tasks).

2. **Ten composable inference primitives.** SelKV, SMSA, Delta-AR, ΔRank, HeadDeactivate, StateCompress, DenseEvic, Pareto Frontier, MoE Router-Gated Importance, and Gating-Aware KV.

3. **Four novel architectural stabilizers.** Manifold-Preserving State Recycling (MPSR), Entropy-Buffered Autoregression (EB-AR), Curvature-Adaptive Spectral Projection with Neural Delta-Phase Alignment (CASP+NDPA), and Contextual Salience Amortization (CSA/IPSS).

4. **Empirical validation on consumer hardware.** All experiments run on a single RTX 4060 (8 GB VRAM).

2 Signal Extraction Framework

OpusEdge resolves the architecture at load time and activates the appropriate signal extraction mode. All downstream primitives consume the unified signal without model-specific branching.

2.1 Native Δ: Hybrid SSM–Attention

In hybrid SSM–attention models, the SSM block maintains a hidden state that evolves as

$$h_{t+1} = \bar{A}\, h_t + \bar{B}\, x_t, \tag{1}$$

where the discrete-time Mamba-2 selective update [3] is

$$\bar{A} = \exp(\Delta_t A), \qquad \bar{B} = \Delta_t B. \tag{2}$$

The information gain from token t is proportional to Δ_t, and the SSD mask [3] has entries

$$M_{ij} = \exp\left(- \sum_{k=j+1}^{i} \Delta_k \right), \tag{3}$$

which is structurally identical to a causal attention matrix with exponential decay.

2.2 Δ_{proxy}: Dense Models

Dense models emit no native Δ. We compute

$$\Delta_{\text{proxy}\,t} \;=\; \frac{1}{L\sqrt{d}} \sum_{\ell=1}^{L} \|h_{\ell,t} - h_{\ell,t-1}\|_2, \tag{4}$$

where $h_{\ell,t}$ is the hidden state at layer ℓ and timestep t. The $\sqrt{d}$ denominator guarantees scale-independence from the model's hidden width, preventing overflow in low-precision regimes.

2.3 Router-Gated Importance IR: MoE

For sparse MoE architectures, the router softmax outputs encode a token-specific importance signal. Given gating weights g_e across E experts, we define two complementary formulations:

$$\mathrm{IR}_t^{\text{maxmin}} = \max_e g_{e,t} - \min_e g_{e,t}, \tag{5}$$

$$\mathrm{IR}_t^{\text{ent}} = 1 - \frac{H(g_t)}{\log E}. \tag{6}$$

3 The Ten Primitives

3.1 SelKV: Zero-Cost KV Cache Eviction

SelKV evicts tokens from the KV cache by zeroing their attention scores. Given importance scores (Δ, Δ_{proxy}, or attention-based), the bottom-p% of tokens are masked. The pipeline is: (1) run SSM layers first (cheaper); extract Δ per head at zero cost; (2) compute token importance Δ_t; (3) append to the KV cache and evict the lowest-Δ entries if over budget; (4) run attention on the pruned cache.

3.2 SMSA: SSM-Masked Sparse Attention

In parallel hybrid blocks, the SSM head provides implicit long-range coverage via the mask M of Eq. 3. The attention head redundantly recomputes these dependencies at $\mathcal{O}(S^2)$ cost. SMSA replaces full attention with an adaptive sliding-window attention of width $w(\Delta)$. At w=64, the KV cache stores only w entries instead of the full sequence, yielding 96.9 % VRAM reduction at 2 048 tokens (Table 2).

3.3 Delta-AR: Δ-Guided Attention Routing

Delta-AR routes each query to the top-K historical tokens by Δ *before* the attention kernel runs. Attention FLOPs drop from $\mathcal{O}(S^2D)$ to $\mathcal{O}(SKD)$ with $K \ll S$. With EB-DAR (Section 4), tokens masked from routing contribute their delta residual to a temporal reservoir rather than being silently dropped.

3.4 ΔRank: Selectivity-Adaptive Low-Rank

High-Δ tokens require full multi-head expressivity; low-Δ tokens can use a rank-r approximation. We use a four-tier mapping (ranks 16/32/64/128 at Δ thresholds); reducing rank from 128 to 32 yields 16× FLOP reduction on the Q/K/V matmul. Combined with SSR (Section 4), the low-rank error is bounded exactly where rank reduction is applied.

3.5 HeadDeactivate: Adaptive Multi-Head Gating

HeadDeactivate gates heads by per-token informational entropy via a four-tier active-head budget

Table 1: Applicability of the ten core primitives across architecture families. Per §1 contribution 2, every primitive is universal; the underlying signal differs.

Primitive	**Dense** (Δ_{proxy})	**Hybrid** (native Δ)	**MoE** ($\Delta \oplus$IR)
SelKV / DenseEvic	✓	✓	✓
SMSA	✓	✓(primary)	✓
Delta-AR + EB-DAR	✓	✓	✓
ΔRank	✓	✓	✓
HeadDeactivate	✓	✓	✓
StateCompress	✓	✓(primary)	✓
GAKV / R-GAKV	✓	✓	✓(primary)
Pareto Frontier	✓	✓	✓
Router-IR	–	–	✓
Stabilizer – MPSR/SACT	–	✓	–
Stabilizer – NDPA	✓	–	–
Stabilizer – SSR/CASP	✓	–	–
Controller – CAL	✓	–	✓
Controller – R-CAL	–	✓	–

Table 2: KV cache memory reduction from SMSA with w=64.

Context	Baseline KV	SMSA KV	Reduction
128	4.7 MB	2.4 MB	50.0 %
256	9.4 MB	2.4 MB	75.0 %
512	18.9 MB	2.4 MB	87.5 %
1024	37.8 MB	2.4 MB	93.8 %
2048	75.5 MB	2.4 MB	**96.9 %**

(4/16/24/32 heads on a 32-head model). Sub-threshold heads fall through to the IPSS linear-time path rather than being zeroed, preventing hard quality cliffs.

3.6 StateCompress: Hidden-State Dimension Compression

When $\Delta \ll \tau$ the SSM state update collapses into passive exponential decay. StateCompress zeros the bottom-$(1-\gamma)$% of channels by magnitude; the keep-ratio $\gamma(\Delta)$ is a monotonically non-decreasing function of Δ.

3.7 DenseEvic: Dense-Architecture KV Eviction Baseline

DenseEvic is the dense-architecture counterpart to SelKV: it accumulates Δ_{proxy} scores in a persistent runtime register, partitions the cache into a protected boundary $\mathcal{B}$ and a candidate pool $\mathcal{C}$, and evicts from $\mathcal{C}$ when the cache exceeds the VRAM budget.

3.8 Pareto Frontier: Multi-Objective Optimization

Compute C, memory M, and quality Q form a conflicting triad. The frontier is computed offline over the discrete configuration grid and used as the system control layer, guiding remaining primitives to the optimal operating point for the current thermal / battery envelope.

3.9 GAKV: Gating-Aware KV Eviction

GAKV integrates Δ and MoE router confidence into a composite retention score $S_t = \alpha\Delta_t + \beta\text{IR}_t$, evicting when $S_t < \tau$. R-GAKV multiplies the thresholds by task rigidity (CAL, Section 5).

4 Stabilizers

Four small primitives prevent quality cliffs under aggressive compression.

MPSR / SACT. Rather than discarding evicted KV tensors, MPSR projects them into the recurrent state via a lightweight linear mapping, constructing an infinite-horizon lossy overflow buffer.

EB-AR / EB-DAR. EB-AR scales compute per step by the Shannon entropy of the output distribution; EB-DAR maintains a temporal delta-error reservoir so tokens masked from Delta-AR routing contribute their residual at subsequent steps.

SSR / CASP + NDPA. SSR replaces hard SVD truncation with a sigmoid-gated soft threshold at the k-th singular value; CASP is the curvature-adaptive variant. NDPA is a rank-1 cross-attention rectifier that aligns Δ_{proxy} onto the internal attention manifold via a single least-squares step at inference time.

IPSS / CSA. When head salience s_h falls below threshold, the quadratic attention is replaced by a linear-time K-averaged fallback that continues projecting into the residual stream without incurring quadratic cost.

5 Task-Adaptive Locking (CAL / R-CAL)

Aggressive compute reduction risks breaking chain-of-thought integrity. CAL dynamically modulates primitive thresholds by classifying prompts into rigidity tiers:

$$\tau_{\text{eff}} = \tau\big(1 + \rho(\text{task})\big), \qquad (7)$$

where ρ encodes domain rigidity (e.g. 0.85 for code, 0.30 for summarization). R-CAL adds an EMA over token log-prob confidence and freezes recomputation once the model has settled.

6 Experimental Setup

Hardware. NVIDIA RTX 4060 (8.2 GB VRAM), 32 GB RAM, Pop!_OS Linux; PyTorch 2.12.1, CUDA 12.4, bitsandbytes 0.43, FP16 compute.

Models. Falcon-H1-0.5B-Instruct [4] (hybrid, 36 layers, WT-103 PPL=11.17); Qwen2.5-0.5B-Instruct [12] (dense, 48 layers, PPL=12.83); AI21-Jamba-Reasoning-3B [9] (52 SSM + 2 attn layers, PPL=7.80); IBM Granite-3.1-1B-A400M (MoE, 1.3 B total / 400 M active).

Metrics. Perplexity via cross-entropy on WT-103 test sequences. Spearman rank correlation. Speedup ratio vs full attention. VRAM footprint. Forward latency in ms with warmup 5, iterations 20.

7 Results

7.1 Δ–Attention Correlation

Falcon-H1 achieves mean ρ=0.51 with 10/24 tasks reaching $p < 0.05$; per-layer analysis confirms 34/36 SSM layers exhibit positive ρ between Δ and attention importance. Qwen2.5 achieves mean ρ=0.28 with 8/24 tasks significant. The gap reflects the informational advantage of a learned selectivity gate over a drift-based proxy.

7.2 SelKV Eviction Quality

Table 3 shows the SelKV quality advantage of 100.5× over random eviction at 87.5 % compression. The regime comparison at 87.5 % across architectures (near-identical ratios: 13.0 hybrid vs 13.7 dense) confirms model-agnostic quality retention.

Table 3: SelKV eviction on Falcon-H1-0.5B (WT-103 long-context, 227 tokens). Baseline PPL= 1.30.

Eviction	SelKV PPL	Random PPL	Ratio
25 %	1.40	11.99	8.6×
50 %	1.84	55.61	30.3×
62.5 %	1.99	144.90	72.8×
75 %	3.63	491.50	135.3×
87.5 %	5.16	519.01	**100.5**×
93.8 %	18.55	1923.84	103.7×

Table 4: SMSA measured speedup (w=64).

Model	Seq	Full (ms)	SW (ms)	Speedup
Falcon-H1	512	0.027	0.013	2.06×
Falcon-H1	2048	0.126	0.035	**3.56**×
Qwen2.5	512	0.011	0.011	1.04×
Qwen2.5	2048	0.083	0.017	**4.98**×

7.3 SMSA Speedup and Memory

Table 4 reports measured SMSA speedup. Qwen2.5 exceeds the roofline 2.8× bound because its smaller per-head dimension amplifies the sliding-window benefit under eager execution.

7.4 Integrated Ablation

Table 5 shows 98.9 % VRAM reduction at 2 048 tokens with the full primitive stack.

7.5 Reference Implementation Verification

The accompanying reference implementation at github.com/Mr-DS-ML-85/OpusEdge verifies the paper's claims empirically on real HuggingFace models. Under a *chunked-SelKV* inference mode that trims the cache to a fixed budget between chunks, we stream 65 536 tokens through Qwen2.5-0.5B (dense) and IBM Granite-3.1-1B-A400M (MoE) at a *measured* 93.8 % KV-cache reduction on peak 1,805 MiB and 3,124 MiB VRAM respectively; the Falcon-H1-0.5B hybrid reaches 16 384 tokens under the same budget. The reduction is computed by summing $K\cdot$numel $+ V\cdot$numel across every cache layer immediately before and after each eviction, so no number is analytical.

8 Related Work

KV cache eviction. H_2O [15] identifies attention sinks and heavy-hitter tokens. SAGE-KV [2] adds frequency and recency scoring. Ada-KV [14] introduces per-layer adaptive budgets. All compute token importance from attention scores at $\mathcal{O}(S)$–$\mathcal{O}(S^2)$ cost, redundantly reconstructing information that hybrid models produce for free via Δ.

Table 5: Cumulative KV memory reduction on Falcon-H1-0.5B.

Context	Baseline	Full pipeline (reduction)
128	5.3 MB	0.7 MB (87.5 %)
512	18.9 MB	0.8 MB (95.8 %)
2048	75.5 MB	0.8 MB (**98.9 %**)

Hybrid SSM–attention. Mamba [5] introduced selective SSMs with input-dependent gating; Mamba-2 [3] proved the SSD theorem. Falcon-H1 [4] uses a 1:1 parallel design; Jamba [9] employs a 1:7 attention-to-Mamba ratio.

Sparse attention. Longformer and BigBird apply fixed sparsity patterns without content adaptation. None exploit the SSM Δ signal as a routing oracle.

Dynamic inference. Early-exit [13], token merging [1], and speculative decoding [8] reduce compute at the sequence or layer level and compose with OpusEdge primitives.

MoE. Mixtral [6] and OLMoE [11] produce routing logits that OpusEdge harvests as the IR signal.

9 Limitations and Future Work

Measured SMSA speedup reflects PyTorch eager-mode prototypes; custom fused CUDA kernels are required to realize the roofline I/O bound at all sequence lengths. Primary evaluation targets the 0.5 B–3 B parameter range; scaling to 7 B+ requires quantized multi-GPU pipelines beyond the current single-GPU harness. Full MPSR effectiveness requires training the projection matrix; the current implementation uses an orthogonal-projection surrogate. Future directions include Δ-guided dynamic precision and cross-layer Δ propagation via a recurrent budget controller.

10 Conclusion

We presented OpusEdge, an end-to-end dynamic compute allocation framework for dense, MoE, and hybrid SSM–attention inference. Harvesting the native Δ in hybrid models, its RMS-drift proxy in dense models, and the router-gated importance signal in MoE architectures, OpusEdge orchestrates ten composable primitives across memory, routing, and conditional computation without retraining. Four novel stabilizers (MPSR, EB-AR, CASP+NDPA, CSA/IPSS) prevent information loss under aggressive compute reduction. SelKV achieves a 100.5× quality advantage at 87.5 % KV compression; SMSA achieves 3.56–4.98× measured attention speedup; StateCompress delivers 37.5 % hidden-state compression with near-zero perplexity degradation. The framework requires no retraining, is hardware-agnostic, and runs on a single consumer GPU—a principled and extensible path toward universal mobile frontier AI.

Code. github.com/Mr-DS-ML-85/OpusEdge

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